

Balancing Trade-offs: Adaptive Differential Privacy in Interpretable Machine Learning Models

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Abstract—In the advancing field of machine learning, balancing accuracy, interpretability, and privacy represents a significant challenge. The problem is exacerbated by the widespread deployment of pre-trained models locally in diverse applications, which could lead to various amounts of privacy leakage. Conventional Differential Privacy strategies, in which uniform noises are applied to model gradients, guarantee data privacy at the expense of accuracy and interpretability. This paper introduces a Feature-Sensitive Adaptive Differential Privacy (FADP) framework with a unique noise-adding strategy. Noises are adaptively added based on feature importance clustering, where important features are considered for interpretability. By employing a unique masking technique, FADP selectively preserves crucial features with minimal noise interference, maintaining accuracy while enhancing interpretability. The FADP framework addresses the limitations of traditional DP methods by preserving critical channels and improving interpretability — a vital requirement in machine learning applications that demand transparency in model decisions. Through comprehensive testing, FADP is shown to balance the trade-offs among accuracy, privacy, and interpretability, marking a substantial advancement in the field of privacy-preserving machine learning.

Index Terms—Privacy-Accuracy-Interpretability Tradeoffs, Differential Privacy, Feature Importance

I. INTRODUCTION

Machine learning (ML) models are increasingly employed across diverse sectors, such as healthcare and finance, where the privacy of sensitive training data is a major concern [1]. These models are prone to memorizing data, exposing them to Membership Inference Attacks (MIA), where adversaries deduce whether specific data points were used in training [2]. As machine learning models are increasingly deployed in real-world applications, the risk of privacy breaches grows, prompting the adoption of Differential Privacy (DP) as a protective measure. DP is typically implemented by adding Gaussian noise to model gradients during training to obscure the contribution of individual data points and protect privacy. Traditional DP strategies are widely known to ensure privacy

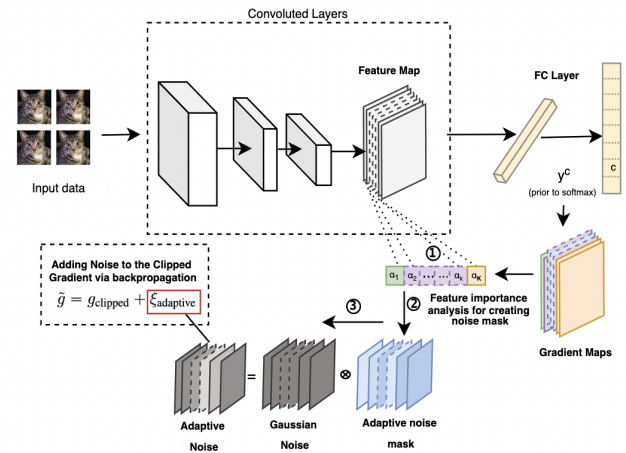


Fig. 1: FADP framework overview

at the cost of reduced model accuracy. In addition to accuracy, there is growing emphasis on model transparency and interpretability [3], especially as machine learning becomes integral to high-stakes applications. However, uniformly applied DP noise can degrade both accuracy and interpretability, intensifying the trade-off among these objectives [4].

We propose a novel framework, *Feature-Sensitive Adaptive Differential Privacy (FADP)*, that selectively applies noise based on feature importance to address the limitations of uniform DP noise. By adaptively scaling noise across features, FADP preserves critical components that contribute most to model interpretability and accuracy. At the same time, because noise is carefully applied to all features and none are left unperturbed, the framework maintains the differential privacy guarantee.

Unlike conventional DP-SGD [5], which uniformly injects noise into all gradients and may obscure meaningful patterns, FADP aligns noise intensity with feature significance. We

utilize feature maps and gradient maps from the last convolutional layer of Convolutional Neural Networks (CNN) to guide adaptive noise mask generation. This layer captures high-level abstractions critical to decision-making, and preserving its key activations under noise is vital for interpretability and accuracy [6].

The key contributions of this work are as follows:

- We introduce FADP, a unique adaptive DP mechanism that applies noise based on feature importance rather than uniformly.
- Our framework bridges the gap between privacy preservation and model interpretability, aiming for a balanced trade-off.
- By clustering feature maps using combined gradient and activation information, we design effective noise masks to selectively reduce perturbation on critical components.
- We provide a systematic method for noise scaling via tunable mask values that control noise intensity and improve overall model utility.
- Empirical results on benchmark datasets demonstrate that FADP significantly improves accuracy and interpretability under privacy constraints compared to standard DP techniques.

II. RELATED WORK

DP is widely adopted to protect individual data contributions in machine learning models by injecting controlled noise into computations [7]. Foundational work by Dwork *et al.* [8] formalized DP, and Abadi *et al.* [9] introduced the DP-SGD algorithm, applying Gaussian noise uniformly to gradients. While effective for privacy, this approach often reduces model accuracy and disregards the role of parameter importance. To mitigate such drawbacks, adaptive DP methods have been proposed [10], [11], dynamically scaling noise based on sensitivity or data properties. However, these efforts have primarily focused on privacy-accuracy trade-offs without considering interpretability.

Interpretability is critical for transparency in high-stakes applications, and techniques like Grad-CAM [12] and SHAP [13] are used to visualize model reasoning. Studies have shown that uniform DP noise severely degrades interpretability [14], highlighting the need for privacy-preserving methods that retain model explainability.

Recent works have begun to address this gap [15]–[18], but many rely only on feature-based interpretation or fail to prevent information leakage via unmasked critical features. Unlike these, our proposed FADP framework combines both feature and gradient-based insights to adaptively scale noise, ensuring stronger privacy guarantees while preserving accuracy and interpretability, particularly for image-based models where gradient locality is key.

III. BACKGROUND AND OVERVIEW

Notations. Variables like x and i denote scalars; $\mathbf{x}$ denotes vectors. X, Y represent datasets and labels; θ , model parameters; σ , noise scale. α_c^k indicates importance weight for feature

map k and class c ; M_k , the adaptive noise mask; $\nabla \mathcal{L}(\theta)$, gradients; $\tilde{\nabla}_k$, clipped gradients.

A. Baseline and MIA Vulnerability

CNNs trained without privacy mechanisms often memorize training data, making them vulnerable to MIA. Given training inputs $X = \{x_1, \dots, x_n\}$ with labels $Y = \{y_1, \dots, y_n\}$, the model f_θ is optimized to minimize the loss $\mathcal{L}(f_\theta(X), Y)$. Overfitting can cause the model to assign overly confident predictions to training points, which adversaries can exploit.

A typical MIA infers membership using model confidence:

$$\hat{m}_i = \begin{cases} 1 & \text{if } p_\theta(x_i) > \tau \\ 0 & \text{otherwise} \end{cases}$$

where $p_\theta(x_i) = \text{softmax}(f_\theta(x_i))$, and τ is a threshold chosen by the adversary. Data points with higher confidence scores are more likely to be identified as part of the training set.

B. Benchmark DP: DP-SGD

DP-SGD protects against MIAs by injecting noise into clipped gradients during training, thereby limiting the influence of any individual data point. For a mini-batch X_t , gradients are first clipped to a maximum norm to bound sensitivity:

$$\tilde{\nabla} \leftarrow \nabla \cdot \max\left(1, \frac{\|\nabla\|}{C}\right) \quad (1)$$

Then, Gaussian noise is added to the clipped gradients to obscure exact contributions from individual samples:

$$\theta_{t+1} = \theta_t - \eta \left(\tilde{\nabla} + N(0, \sigma^2) \right) \quad (2)$$

The noise scale σ is calibrated using the privacy parameters ϵ , δ , and sensitivity Δ as follows:

$$\sigma = \frac{\Delta}{\epsilon} \cdot \sqrt{2 \log \frac{1.25}{\delta}} \quad (3)$$

This ensures the training process satisfies (ϵ, δ) -differential privacy. Larger ϵ implies less noise (weaker privacy, higher utility), and vice versa. Though effective, this uniform noise addition can degrade both accuracy and interpretability.

C. Interpretability with Grad-CAM

Grad-CAM provides class-discriminative visual explanations by using gradients with respect to feature maps. Let $A_k(x)$ be the k -th activation map and y_c the class score. Importance weights are:

$$\alpha_{ck} = \frac{1}{Z} \sum_i \sum_j \frac{\partial y_c}{\partial A_{ij}^k(x)} \quad (4)$$

The Grad-CAM heatmap is:

$$L_{\text{Grad-CAM}}^c(x) = \text{ReLU} \left(\sum_k \alpha_{ck} A_k(x) \right) \quad (5)$$

These heatmaps highlight regions contributing most to the prediction. However, DP noise can distort these explanations, weakening model transparency—an issue FADP directly addresses.

D. Feature-Sensitive Adaptive Differential Privacy (FADP)

FADP aims to balance privacy, accuracy, and interpretability by introducing adaptive noise based on feature importance. Unlike DP-SGD, which applies uniform noise to all parameters, FADP assigns lower noise to critical features and stronger noise to less important ones, preserving interpretability without sacrificing privacy.

Feature importance is computed using the gradient of the output score y_c with respect to the feature maps $A_k(x)$, inspired by Grad-CAM:

$$\alpha_{ck} = \frac{1}{Z} \sum_i \sum_j \frac{\partial y_c}{\partial A_{ij}^k(x)} \quad (6)$$

These weights are clustered into high, moderate, and low importance. Adaptive noise masks M_k are assigned accordingly:

$$M_k = \begin{cases} m_{\text{low}} = 1 & \text{low importance} \\ m_{\text{moderate}} & \text{moderate importance} \\ m_{\text{high}} < 1 & \text{high importance} \end{cases}$$

The scaled Gaussian noise $M_k \cdot \mathcal{N}(0, \sigma^2)$ is then added to the clipped gradient $\tilde{\nabla}_k$, where:

$$\tilde{\nabla}_k = \nabla_k + M_k \cdot \mathcal{N}(0, \sigma^2) \quad (7)$$

This ensures that important features receive less perturbation, maintaining decision-critical activations, while unimportant ones receive more noise for privacy protection.

FADP Framework Overview (Figure 1):

- 1) Extract feature maps and gradients for each input.
- 2) Compute importance scores α_{ck} and cluster them into three importance levels.
- 3) Generate Gaussian noise and modulate it using the adaptive mask M_k .
- 4) Add masked noise to the clipped gradient during back-propagation.

Algorithm 1 summarizes this process. Each iteration computes importance weights, assigns masks, clips gradients, applies noise, and updates model parameters. The clustering into three tiers smooths noise distribution and avoids revealing identifiable noise patterns.

Algorithm 1 FADP: Adaptive DP with Feature Clustering

- 1: **Input:** Dataset X , Learning rate η , Noise scale σ , Clipping norm C
 - 2: **for** each iteration **do**
 - 3: Compute gradients $\nabla \mathcal{L}(\theta_t, X_t)$
 - 4: Compute importance weights α_{ck} , cluster into 3 groups
 - 5: Assign adaptive noise mask M_k to each feature map
 - 6: Clip gradients: $\tilde{\nabla}_k \leftarrow \nabla_k / \max(1, \frac{\|\nabla_k\|}{C})$
 - 7: Add adaptive noise: $\tilde{\nabla}_k \leftarrow \tilde{\nabla}_k + M_k \cdot \mathcal{N}(0, \sigma^2)$
 - 8: Update model: $\theta_{t+1} \leftarrow \theta_t - \eta \tilde{\nabla}_k$
 - 9: **end for**
-

Privacy Guarantee : The FADP framework satisfies (ϵ, δ) -differential privacy by adaptively scaling Gaussian noise based

on feature importance. For each feature i , noise is added to its gradient as:

$$g'_i(\theta) = g_i(\theta) + \mathcal{N}(0, \sigma_i^2), \quad \text{where } \sigma_i(\alpha_i) = \frac{\Delta_i}{\epsilon} \sqrt{2 \log \frac{1.25}{\delta}}$$

Here, Δ_i is the sensitivity of feature i , which is inversely related to its importance score α_i . This ensures that more important features receive lower noise to preserve model utility, while less important features are perturbed more to reinforce privacy. The mechanism guarantees that for any two adjacent datasets D and D' differing by one sample:

$$\mathbb{P}[f_\theta(D) \in S] \leq e^\epsilon \mathbb{P}[f_\theta(D') \in S] + \delta$$

This satisfies the formal definition of differential privacy while allowing for selective preservation of interpretability.

IV. THREAT MODEL

The FADP framework is designed to defend against black-box Membership Inference Attacks (MIA), where the adversary cannot access model parameters but can observe output predictions.

A. Adversary Model

We assume a black-box adversary who queries the model with selected inputs and observes output probabilities [19]. Has some knowledge of the data distribution [20]. Operates under a query budget, as in practical settings [21].

B. Attack Vector and Defense Mechanism

The main attack considered is MIA, where the adversary aims to distinguish whether a data point was part of the training set. FADP defends against this by Obscuring Patterns as the adaptive noise disrupts confidence patterns linked to training samples. The mask variability hinders accurate shadow model replication.

FADP ensures that for adjacent datasets X and X' , the model outputs remain statistically indistinguishable:

$$\mathbb{P}(f_\theta(X) = o) \approx \mathbb{P}(f_\theta(X') = o)$$

as bounded by DP parameters (ϵ, δ) .

C. Security Guarantees

FADP provides a twofold defense: leftmargin=1.2em

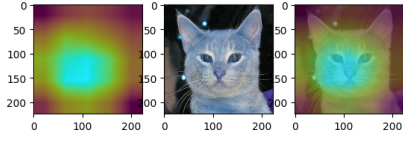
- **Gradient Clipping:** Limits individual data influence.
- **Adaptive Noise Scaling:** Adds stronger noise to less important features, reducing MIA risk while preserving utility.

This ensures effective privacy protection while maintaining accuracy and interpretability.

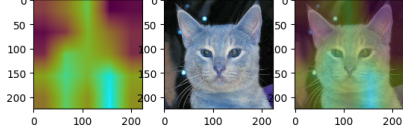
V. EVALUATION

A. Experimental Setup

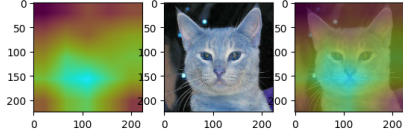
We evaluate FADP on its ability to balance privacy, accuracy, and interpretability using both visual and numerical metrics.



(a) Grad-CAM output for Baseline Model (No DP)



(b) Grad-CAM output for Benchmark DP (DP-SGD)



(c) Grad-CAM output for the proposed FADP

Fig. 2: Grad-CAM heatmaps for different models to demonstrate model interpretability

1) **Environment and Tools:** All experiments were conducted on Google Colab TPU (T4) using TensorFlow (Python 3.x). NumPy, Matplotlib, OpenCV, and scikit-learn were used for preprocessing, visualization, and evaluation.

2) **Model Architecture:** We use **MobileNet** with ImageNet pre-trained weights [22], enhanced with an additional convolutional layer and fully connected layers to support Grad-CAM-based feature importance estimation for FADP.

3) **Datasets:** We evaluate the FADP framework on two datasets. The **CIFAR-10** dataset [23] consists of 60,000 32×32 color images across 10 object categories, with 50,000 training and 10,000 test samples. The **Cat vs. Dog** dataset [24], obtained from Kaggle, contains 25,000 labeled images evenly split between cats and dogs. These datasets enable us to assess the scalability of FADP across both multi-class and binary classification tasks. The latter is particularly valuable, as binary models are more susceptible to overfitting and membership inference attacks [25].

4) **Evaluation Metrics:** We assess classification performance using accuracy and loss on the test set. Privacy leakage is evaluated through MIA, using the AUC score with confidence intervals to measure the attacker's success. For interpretability, we analyze Grad-CAM heatmap overlays and quantitatively compare them using Structural Similarity Index (SSIM) [26] and Peak Region Overlap to assess how well the model maintains focus on critical features under different privacy settings.

5) **Training Setup and Hyperparameters:** The model was trained using the Adam optimizer with a learning rate $\eta = 0.001$, batch size 64, and clipping norm $C = 1.0$. FADP showed stable performance across common batch sizes and clipping values. To satisfy privacy guarantees, we set $\epsilon \in$

Phase	Baseline		DP-SGD		FADP	
	Acc.	Loss	Acc.	Loss	Acc.	Loss
Train	0.9215	0.2310	0.8125	0.7063	0.882	0.3013
Val.	0.9110	0.2635	0.8200	0.4124	0.8750	0.2896
Test	0.8701	0.4981	0.8250	0.3584	0.8500	0.4286

TABLE I: Comparison of Baseline, DP-SGD, and FADP Models on Cat and Dog Dataset

Phase	Baseline		DP-SGD		FADP	
	Acc.	Loss	Acc.	Loss	Acc.	Loss
Train	0.9310	0.2100	0.8100	0.6520	0.8600	0.2031
Val.	0.9200	0.2320	0.8250	0.3990	0.8650	0.2790
Test	0.8850	0.2590	0.8350	0.2910	0.8400	0.2390

TABLE II: Comparison of Baseline, DP-SGD, and FADP Models on CIFAR-10 Dataset

$[0.1, 3]$ with $\delta = 10^{-5}$, ensuring noise levels $\sigma \in [\sigma_{\min}, \sigma_{\text{base}}]$ as per equation 3. Adaptive noise scaling was applied using $m_{\text{high}} \approx 0.6$, $m_{\text{moderate}} \approx 0.8$, and $m_{\text{low}} = 1$, allowing reduced noise on important features while preserving DP bounds.

6) **Model Comparisons:** For comparison, we evaluated the following models in each result section:

- **Baseline Model (No DP):** A model trained without any privacy-preserving mechanisms.
- **DP-SGD (Benchmark DP):** A model trained using Stochastic Gradient Descent Differential privacy with uniform noise addition after gradient clipping.
- **FADP:** A model trained using the proposed FADP technique with adaptive noise.

Model	Input	Baseline	Compared	SSIM	Peak Overlap
DP-SGD				0.31	14%
				0.45	23%
Proposed FADP				0.86	81%
				0.81	83%

TABLE III: SSIM and Peak Region Overlap score comparison with baseline model

Model	Metric	Avg.Score	Standard Deviation
Benchmark DP-SGD	SSIM	0.42	0.035
	Peak Overlap	30%	6%
Proposed FADP	SSIM	0.85	0.033
	Peak Overlap	83%	5%

TABLE IV: Summary of average SSIM and Peak Region Overlap for 100 samples across models.

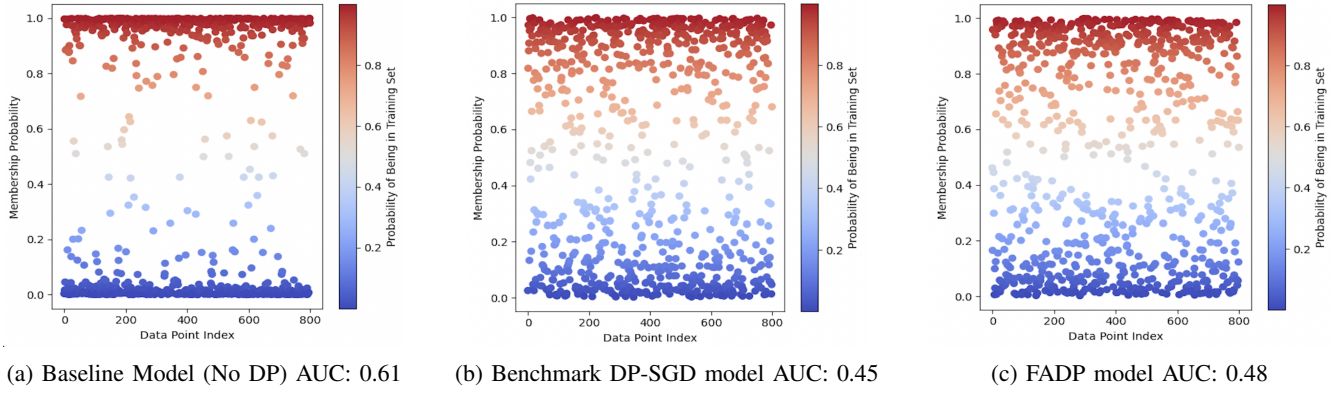


Fig. 3: Comparison of AUC Scores of the MIA Attack models on the Baseline, Benchmark DP (DP-SGD), and FADP Models

Case	Model Parameters	Accuracy (%)		MIA Attack AUC ($\pm$ CI)	
		Cat vs Dog	CIFAR-10	Cat vs Dog	CIFAR-10
Case 1	$m_{\text{high}} = 0.5, m_{\text{moderate}} = 0.7, m_{\text{low}} = 1$	89.50	87.50	0.51 ± 0.05	0.54 ± 0.03
Case 2	$m_{\text{high}} = 0.55, m_{\text{moderate}} = 0.75, m_{\text{low}} = 1$	89.00	86.00	0.49 ± 0.04	0.54 ± 0.03
Case 3	$m_{\text{high}} = 0.6, m_{\text{moderate}} = 0.8, m_{\text{low}} = 1$	88.00	84.00	0.45 ± 0.06	0.48 ± 0.05
Case 4	$m_{\text{high}} = 0.65, m_{\text{moderate}} = 0.85, m_{\text{low}} = 1$	86.50	81.50	0.44 ± 0.05	0.47 ± 0.04

TABLE V: FADP Model Accuracy and MIA Attack AUC Scores with CI for Different m Values

B. Results

1) **Accuracy Improvement Analysis:** The FADP framework significantly improves accuracy over DP-SGD, as shown in Table I on the Cat and Dog Dataset. The Baseline model achieved 92.15%, 91.10%, and 87.01% accuracy on training, validation, and testing, respectively. DP-SGD dropped to 81.25% training accuracy, highlighting the privacy-utility trade-off. FADP improved this to 88.2%, with validation and testing accuracies of 87.50% and 85.00%, nearly matching the Baseline. On CIFAR-10 (Table II), FADP also outperformed DP-SGD with 86.00% training, 86.50% validation, and 84.00% testing accuracy, compared to DP-SGD's 81.00% training.

2) **Interpretability Improvement Analysis:** Figure 2 compares the interpretability of the Baseline (No DP), DP-SGD, and FADP models using Grad-CAM heatmaps to highlight prediction-relevant regions. In Figure 2a, the Baseline model sharply focuses on facial features like the eyes and nose, indicating high interpretability. In contrast, Figure 2b shows that DP-SGD produces diffused heatmaps, spreading focus to irrelevant areas due to uniform noise. Figure 2c demonstrates that FADP retains attention on key regions. Although slightly broader than the baseline, the adaptive noise mechanism preserves decision-critical focus areas. Table III presents SSIM and peak overlap scores relative to the Baseline. DP-SGD's lower SSIM values (0.31, 0.45) reflect decreased alignment, while FADP achieves higher scores (0.86, 0.81), indicating better preservation of focus. Peak overlap results follow a similar trend. Additionally, Table IV summarizes average SSIM and Peak Region Overlap across 100 samples with standard deviations, confirming the consistency of FADP's interpretability. These results highlight FADP's ability to bal-

ance privacy and interpretability by adapting noise based on feature importance. Unlike DP-SGD, which scatters model attention, FADP emphasizes critical regions, demonstrating its effectiveness in maintaining transparency under privacy constraints.

3) **Under Attack Performance Analysis:** Figure 3 presents MIA results for the baseline, DP-SGD, and FADP models using heatmaps that indicate membership probabilities for the Cat and Dog datasets. In Figure 3a, the baseline model shows a clear separation between training and non-training data, with membership probabilities close to 1 for training data and near 0 for non-training data. This distinct separation, reflected by an AUC score of 0.59, highlights the model's vulnerability to MIA. In contrast, Figure 3b shows that DP-SGD introduces noise that blurs this distinction, resulting in overlapping membership probabilities and a reduced AUC score of 0.45. While this improves privacy, it comes at some cost to model performance. Figure 3c illustrates the FADP framework, where membership probabilities are more evenly distributed than in the baseline but more structured than in DP-SGD. With an AUC score of 0.48, FADP offers a better balance between privacy and accuracy, reducing vulnerability while maintaining focused membership distribution. Overall, Figure 3 demonstrates that while the baseline is highly vulnerable to MIA, DP-SGD enhances privacy at the cost of interpretability, whereas FADP achieves a balanced trade-off, reducing attack success while preserving model focus.

4) **Selection of Noise Mask Values:** In Table V, we evaluate the FADP model's performance under varying m values, analyzing both accuracy and the AUC scores of MIA. For each configuration, accuracy and MIA AUC scores are presented. Notably, AUC scores include CI to demonstrate the statistical reliability of our findings. Case 3 ($m_{\text{high}} = 0.6, m_{\text{moderate}} =$

0.8, $m_{\text{low}} = 1$) stands out as particularly effective, providing a well-balanced trade-off between accuracy and privacy. It achieves AUC scores of 0.45 ± 0.06 for Cat vs. Dog and 0.48 ± 0.05 for CIFAR-10, indicating stronger privacy protection compared to other cases. Although the accuracy for Case 3 is slightly lower than in Case 1, at 88.00% for Cat vs. Dog and 84.00% for CIFAR-10, the reduction in AUC underscores a significant improvement in privacy measures. The inclusion of CIs further validates the consistency and robustness of these results across multiple experimental iterations.

VI. CONCLUSION

The proposed framework successfully addresses the critical trade-offs between privacy, accuracy, and interpretability in machine learning models. By applying noise adaptively based on feature importance, rather than uniformly, the FADP model preserves key features essential for decision-making while maintaining adequate privacy. Extensive testing on various datasets demonstrates that FADP achieves a more balanced trade-off, improving interpretability and maintaining high model accuracy while offering strong privacy guarantees. Although the framework incurs slightly higher computational costs, particularly in comparison to standard DP methods, the significant improvements in model performance and privacy preservation underscore the effectiveness of the FADP approach.

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